

Release Note: Lab Infrastructure Modernization

Date: May 2026

Version: 2.0.0 (Unified Stack Migration)

Classification: Internal Lab Operations

1. Executive Summary

This release documents the decommissioning of a heterogeneous "best-of-breed" network and the implementation of a unified **Ubiquiti UniFi Ultra** ecosystem. The architecture has been redesigned to decouple the lab environment from the primary production network, ensuring "break-fix" testing capabilities without sacrificing external connectivity for troubleshooting.

2. Physical Topology & Backhaul

The new design utilizes a distributed switching model connected via a high-bandwidth backbone.

- **Core Gateway (Cloud Gateway Ultra):** Serves as the WAN edge and Site Controller.
- **Lab Distribution (16-Port PoE Switch):** Located in the main lab area, connected to the Gateway via a **dedicated long-haul trunk cable**. This switch acts as the primary power and data hub for lab servers and the UK-Ultra WAP.
- **Workstation Edge (8-Port PoE Switch):** Situated at the desk to provide localized connectivity for "at-hand" testing devices, linked back to the core fabric to maintain VLAN consistency at the user interface.

3. Logical Network Design (Segmentation)

The network has migrated from a flat topology to an **802.1Q VLAN-segmented architecture**. The long-haul cable from the Gateway to the 16-port switch is configured as a **Trunk Port**, carrying all four defined VLANs:

VLAN ID	Name	Description	Purpose
10	Trusted	Primary management & personal devices.	Secure, high-priority traffic.
20	IoT	Low-trust smart devices.	Isolation from core lab assets.
30	Lab	Active development & testing zone.	General lab work/virtualization.
99	Sandbox	High-risk testing environment.	Full isolation for "break-fix" scenarios.

4. Key Technical Improvements

Hardware-Abstracted Management

By migrating the UniFi Controller from a ZimaOS-hosted container to the **dedicated Cloud Gateway Ultra**, the management layer is no longer dependent on a computer's health. This ensures the network dashboard remains accessible even during total lab server failure.

Backbone Reliability

The transition from a mix of QNAP fiber and Flex Mini switches to a unified UniFi trunk minimizes protocol mismatch. The long-haul link between the Gateway and the 16-port switch provides a stable, simplified backhaul that is easier to monitor for CRC errors or throughput bottlenecks.

Redundant Power Efficiency

The 16-port (Lab) and 8-port (Desk) switches provide localized PoE, eliminating the need for power injectors. This allows for:

- **Remote Power Cycling:** Resetting the UK-Ultra WAP or lab nodes via the UniFi app.
- **Clean Desk Policy:** Significant reduction in cable clutter and "wall-wart" power supplies at the workstation.

5. Resilient Troubleshooting Workflow

Because the Gateway manages the WAN independently of the Lab switch, a configuration failure in **VLAN 99 (Sandbox)** does not impact **VLAN 10 (Trusted)**. This allows the user to maintain internet access to research fixes while the lab environment is "broken."

Hardware Inventory & Placement

Model	Location	Physical Interface
Cloud Gateway Ultra	Main Entry	WAN In / Trunk Out
16-Port PoE Switch	Lab	Trunk In (Long-haul) /Distribution
8-Port PoE Switch	Desk	Local Access / Peripheral Testing
UK-Ultra WAP	Lab	Wireless Access (PoE Powered)

Note for Portfolio: *This document highlights the transition to a structured cabling approach. By utilizing a trunked long-haul connection between the Gateway and the 16-Port switch, the designer demonstrates proficiency in managing physical layer constraints while maintaining logical segmentation across the environment.*

End of Release Note